



ORIGINAL RESEARCH

Modeling Dynamic ACL Loading During Running in Post-ACL Reconstruction Individuals: Implications for Regenerative Engineering

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Abstract

Purpose Neuromuscular deficits in post-anterior cruciate ligament reconstruction (ACLR) individuals result in the adoption of compensatory movement strategies that alter how the knee and ACL are loaded. While it is understood that post-ACLR individuals load the knee and ACL differently than healthy controls, the challenge is quantifying the loading in the ACL as it can provide insight regarding secondary ACL tears. Computational modeling is advantageous because it provides a way to compute the forces in the ACL in individuals during dynamic tasks. Thus, this study sought to employ computational modeling to investigate differences in how the ACL is loaded during running in healthy controls and post-ACLR individuals.

Methods Sixteen post-ACLR individuals and twelve healthy controls performed a running protocol where they were instructed to run at a self-selected speed. Subject-specific computational models were generated for each participant, and joint kinematics, joint kinetics, and muscle and ACL forces were extracted from the simulations.

Results Despite the post-ACLR individuals exhibiting significantly lower peak knee flexion angles and peak knee extensor moments than the controls, the peak ACL forces were marginally higher in post-ACLR individuals than in healthy controls ($p = 0.33$).

Conclusion These findings suggest that secondary ACL tears may be more of a reflection of the challenges faced in mimicking the structural and functional roles of the ACL with transplanted tissues rather than the effects of elevated ligament loading. The knowledge gained from this study will have significant implications for regenerative engineering and the design of more robust tissues.

Lay Summary Individuals after an anterior cruciate ligament reconstruction (ACLR) suffer from diminished muscle function that causes them to adopt new movement patterns. A consequence of these new movement patterns is a change in the forces that are placed on the newly transplanted ACL. This study used computational modeling to compare the forces experienced by the ACL in healthy controls and post-ACLR individuals during running. No differences in ACL forces were measured between the two groups, suggesting that secondary ACL tears may be due to the altered structural integrity of the newly transplanted grafts.

Keywords Ligament forces · Neuromuscular control · Pathological gait · Joint biomechanics

Introduction

Post-anterior cruciate ligament reconstruction (ACLR) individuals are often plagued by neuromuscular deficits that result in altered movement dynamics [1–3]. These neuromuscular deficits contribute to the development of knee

osteoarthritis (OA) and secondary ACL tears that are prevalent within this population. While the development of knee OA represents the long-term consequences of how altered muscle activation dynamics change knee loading, secondary ACL tears are indicative of how muscle activation dynamics influence the forces experienced by the ACL [4–7]. Since secondary ACL tears occur on transplanted grafts, it is valuable to understand if the transplanted ACLs are subjected to higher forces during dynamic tasks in post-ACLR individuals or if the challenges associated with mimicking the structural and functional behavior of the ACL with the transplanted grafts

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lead to tears when the ACL experiences submaximal loading. Therefore, this study investigated differences in the forces experienced by the ACL during running in post-ACLR individuals and healthy controls to better understand how changes in movement dynamics due to neuromuscular imbalances may increase the risk for secondary ACL tears.

Extensive cadaveric work has provided vital foundational information that has aided in the characterization of the forces experienced by the ACL under variable loading conditions (Kanamori et al. 2000; Markolf et al. 1990; Woo et al. 1991). Furthermore, these simulated conditions have successfully identified the knee orientation that is commonly associated with ACL rupture [8–10]. However, it is difficult for these cadaveric studies to simulate ACL loading during dynamic tasks in post-ACLR individuals who are suffering from neuromuscular deficits, yet computational modeling can. Personalized musculoskeletal models have been used to evaluate knee loading during dynamic tasks in both healthy and pathological populations [11–14]. These simulation studies provide insight into the internal structural behavior of joints and ligaments. It has been well established that post-ACLR individuals adopt alternate muscle activation strategies than healthy controls during dynamic tasks leading to reduced knee flexion and elevated knee loading [3, 15, 16]. Due to these adaptive changes, the forces in the ACL are believed to increase. Understanding the forces experienced by the ACL is critical for the field of regenerative engineering as the translational information here can be used to design more robust tissues to withstand the forces during dynamic loading [17–19]. Thus, the personalized simulations served to quantify the impact these changes in movement dynamics have on ACL loading during running.

This study sought to analyze the forces experienced by the ACL in post-ACLR individuals and healthy controls during running. Running is a high-impact task requiring greater neuromuscular demands, meaning differences in ACL loading between healthy controls and post-ACLR individuals with neuromuscular deficits would be magnified. Additionally, the stress on the ACL is lower during running than cutting, therefore providing a safer option for investigating differences within these populations. We hypothesized that the forces exerted on the ACL will be greater in the post-ACLR individuals than in the healthy controls during running. This will suggest that alternate muscle activation strategies post-ACLR individuals adopt in response to neuromuscular dysfunction not only lead to elevated knee loading but that this elevated ACL loading is the cause for the high rates of secondary ACL tears. Plus, this knowledge will indicate the magnitude of the dynamic loads exerted on the newly transplanted grafts.

Methods

Instrumented Gait Analysis

Sixteen post-ACLR individuals (age: 19.5 ± 6.3 years.; mass: 70.1 ± 11.0 kg; height: 1.7 ± 0.1 m, speed: 2.7 ± 0.3 m/s; gender: 8 males, 8 females) and 12 healthy controls (age: 21.2 ± 4.3 years.; mass: 63.3 ± 14.6 kg; height: 1.7 ± 0.1 m, speed: 2.6 ± 0.3 m/s; gender: 6 males, 6 females) performed a running protocol that was approved by the institutional review board. The protocol instructed the participants to run on the split-belt instrumented treadmill (Bertec Corp., Columbus, OH) at a self-selected speed for 5 min to get acclimated to the treadmill. Then, the individuals ran at a self-selected speed during which the data was collected. All the participants were between 18 and 40 years old and wore the same WR662 running shoes (New Balance, Brighton, MA). All post-ACLR individuals had undergone surgery 6 months prior to participating in the study and were cleared to return to sport by a physician. All healthy controls were free of injury at least 6 months prior to participating in the study and had no history of knee surgery.

A 56 retro-reflective marker set was used where the markers were placed on the lower and upper extremities. This setup was based on an established design [3]. Markers were placed on the C7 vertebrae, sternum, and bilaterally on the acromion process, iliac spines, iliac crests, greater trochanters, femoral epicondyles, tibial plateaus, malleoli, and the first and fifth metatarsal heads. Four sets of rigid marker clusters were placed on the right and left thighs and shanks.

After the participants were acclimated to the treadmill, the participants performed the running protocol that involved running at a self-selected speed. Ten strides were collected for each limb, and the stride data was used for the analysis. The marker trajectories were collected at 200 Hz by a 10-camera motion capture system (Motion Analysis Corp., Santa Rosa, USA). Ground reaction force (GRF) data was simultaneously collected at 1200 Hz. The marker and GRF data were filtered at 8 Hz and 35 Hz, respectively, using a fourth-order Butterworth low pass filter.

Subject-Specific Model Generation

Personalized simulations were developed for the healthy controls and post-ACLR individuals using OpenSim, an open-source musculoskeletal software program [11, 12]. First, the three-dimensional, 14-segment, 37 degree-of-freedom, 92 muscle–tendon actuated subject-specific running simulations were generated for the healthy controls and post-ACLR individuals (Fig. 1). The models were actuated by the 92 muscle–tendon units, and an ACL was added to the model using an established approach [20] (Fig. 1). The same angular orientation of the ACL was applied to the models for the healthy and

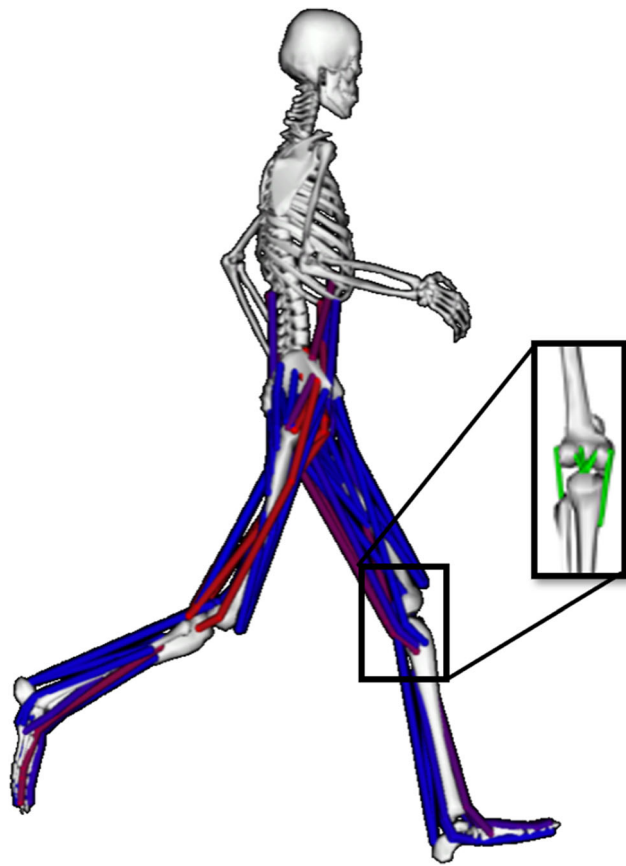


Fig. 1 Representation of muscle-actuated, subject-specific simulation of running. The model has 37 degrees of freedom and is actuated by 92 muscles. The knee contains the anterior cruciate ligament, posterior cruciate ligament, medial collateral ligament, and lateral collateral ligament (inset). Including both the muscle and ligament architecture will enable us to better model the loading dynamics during the walking protocol

post-ACLR individuals [ref]. Each model was then scaled to the participants' segment mass, dimensions, and joint centers. Simulated joint angles were derived using inverse kinematics. Joint kinetics were computed using inverse dynamics, and muscle force data were estimated using static optimization. A custom MATLAB code (MATLAB R2019a, The MathWorks, Inc., Natick, MA) was developed to extract and analyze the peak sagittal plane knee kinematic and kinetic data as well as the muscle force data.

Muscle Activation Dynamics Calculation

Directed co-contraction ratio (DCCR) analysis was employed to evaluate individuals' muscle activation strategies. The total net flexor-to-extensor DCCR muscle activation (FEDCCR) was computed during the braking phase of running. The DCCR is a ratio of the agonist muscles to the antagonist muscles. For FEDCCR, the agonist muscles are the flexor muscles (semimembranosus, biceps femoris, and gastrocnemii), and the antagonist muscles are the extensor muscles (rectus

femoris, vastus lateralis, and vastus medialis). Equations 1 and 2 are used to compute these ratios, and when the ratios are greater than zero, it indicates that the agonist muscles are more dominant than the antagonist muscles [21].

If the agonist mean activation > antagonist mean activation

$$\text{DCCR} = 1 - \frac{\text{agonist mean muscle activation}}{\text{antagonist mean muscle activation}} \quad (1)$$

Else

$$\text{DCCR} = \frac{\text{agonist mean muscle activation}}{\text{antagonist mean muscle activation}} - 1 \quad (2)$$

ACL Force Calculation During Running

The force in the ACL was computed as a function of the change in ACL length and the linear elastic stiffness of ACL. The linear elastic stiffness for the ACL was 240 N/mm, which was obtained from a study by Woo et al. [10]. The ACL force was computed by multiplying the change in length of the ACL from its resting length by 240 N/mm.

Statistical Analysis

A *t* test was conducted to assess differences in mean age, height, mass, and running speed between the control and post-ACLR groups. A *t* test was also performed to evaluate peak sagittal plane knee kinematics and kinetics, FEDCCR, and mean ACL force between the healthy controls and post-ACLR individuals. The Kolmogorov–Smirnov normality test was performed on the study data and supported the use of the *t* test for the statistical analyses. The statistical analyses were conducted in MINITAB (MINITAB, Version 18, State College, PA, USA).

Results

There were no differences in age, height, mass, and running speed between the healthy controls and post-ACLR groups

Table 1 Comparison of participant demographics (mean ± standard deviation)

Variable	Control group	Post-ACLR group	<i>p</i> value
Age (years)	21.2 ± 4.3	19.5 ± 6.3	0.69
Height (m)	1.7 ± 0.1	1.7 ± 0.1	0.15
Mass (kg)	63.3 ± 14.6	70.1 ± 11.0	0.08
Speed (m/s)	2.6 ± 0.3	2.7 ± 0.3	0.80
Gender (female:male)	6:6	8:8	

(Table 1). A comparison of knee kinematics and kinetics revealed that the post-ACLR individuals reported significantly lower peak knee flexion ($37.6 \pm 5.2^\circ$) than the healthy controls ($43.4 \pm 4.9^\circ$) (Fig. 2; Table 2). The post-ACLR individuals also produced significantly lower peak knee kinetics (2.4 ± 0.3 Nm/kg) compared to the healthy controls (1.9 ± 0.4 Nm/kg) (Fig. 2; Table 2). Both the healthy controls and post-ACLR individuals exhibited an extensor muscle dominance during the braking phase of running based on the FEDCCR. The ACL experienced higher forces during running in the post-ACLR individuals (1060.5 ± 335.0 N) compared to the healthy controls (906.7 ± 463.5 N), but it was not significant ($p = 0.33$) (Table 2). Therefore, despite the post-ACLR individuals exhibiting alternate joint motion and muscle activation strategies from the controls, the post-ACLR individuals did not exhibit differences in ACL loading.

Discussion

The objective of this study was to investigate the forces experienced by the ACL in healthy controls and post-ACLR individuals during running. While the post-ACLR individuals did exhibit higher forces in the ACL than the controls, the difference was not significant. The post-ACLR individuals exhibited no significant differences in ACL loading despite generating significantly reduced knee flexion angles and knee extensor moments during running. These movement patterns were adopted in response to the quadriceps dominant strategy exhibited by the post-ACLR individuals. Despite the post-ACLR individuals adopting alternate muscle activation and movement patterns than the controls, these changes did not significantly impact ACL loading. Given the high rates of secondary ACL tears, these findings indicate that secondary tears may not be solely due to elevated forces experienced by the ACL but suggest that transplanted grafts may not have the necessary structural and/or dynamic integrity needed to adapt to the variable loading that can induce ACL failures.

Most ACL tears during sports are noncontact injuries that occur when the forces in the ACL exceed the tissue tolerance [22]. Initial ACL tears are best described as ruptures that occur when the fibers pull apart causing the midsubstance to separate during rapid, high-impact cutting or landing events [23].

However, secondary ACL tears are characterized by different failure mechanisms that include the progression of partial tears into complete tears and failures of the ligament–bone attachment site [5, 23]. The failure mechanisms associated with secondary ACL tears can result in ligament damage under loading conditions when the maximum forces in the ACL are not exceeded. Failures can also occur during these sub-maximal loading conditions because the ligament structure is compromised. While secondary ACL tears have been linked to elevated forces incurred by the ACL due to the compensatory movement patterns adopted by individuals post-ACL reconstruction, the minimally elevated forces measured in the ACL during running suggest that the repeated submaximal loading of the compromised ACL structure may be responsible for the high rates of ACL re-tears.

Computational modeling provides a safe and effective way to analyze the structural behavior of biological materials such as the ACL under dynamic loading. During running, the forces generated in the ACL are the result of a combination of factors such as ACL orientation, material properties, joint motion, and muscle activation dynamics. Here, this study sought to investigate how muscle activation strategies and the associated joint motion impacted the forces experienced by the ACL. Thus, we opted to minimize the impact of ACL orientation by standardizing the ACL orientation across all models given that prior work has shown that ACL loading is sensitive to ACL orientation [24]. By standardizing ACL orientation, it ensured that changes in ACL loading would be attributed to the compensatory movement strategies adopted by the post-ACLR individuals, which will help provide valuable information regarding the risk of secondary ACL tears.

The ACL is a complex structure that functions to dynamically resist anterior tibial translation and rotational motion to ensure knee stability under high loading situations. The dual structural and functional demands of the ACL present significant challenges for new materials implanted to fulfill this role. Tissue and regenerative engineering fields are focused on designing and optimizing novel materials that can outperform the natural materials that they are engineered to replace [17, 18, 25, 26]. The emergence of these fields has coincided with significant advancements in the development of newly bioengineered materials and attachment site anchoring approaches that address current ACL reconstruction challenges

Table 2 Comparison of mean knee joint kinematics, kinetics, muscle activation, and ACL loading (mean $\pm$ standard deviation)

Variable	Control group	Post-ACLR group	<i>p</i> value
Knee flexion (degrees)	43.4 ± 4.9	37.6 ± 5.2	$<0.001^*$
Knee extension moment (Nm/kg)	2.4 ± 0.3	1.9 ± 0.4	0.01^*
FEDCCR	-0.3 ± 0.1	-0.5 ± 0.2	$<0.001^*$
ACL force (Newtons)	906.7 ± 463.5	1060.5 ± 335.0	0.33

*The means were significant between the two groups ($p < 0.05$)

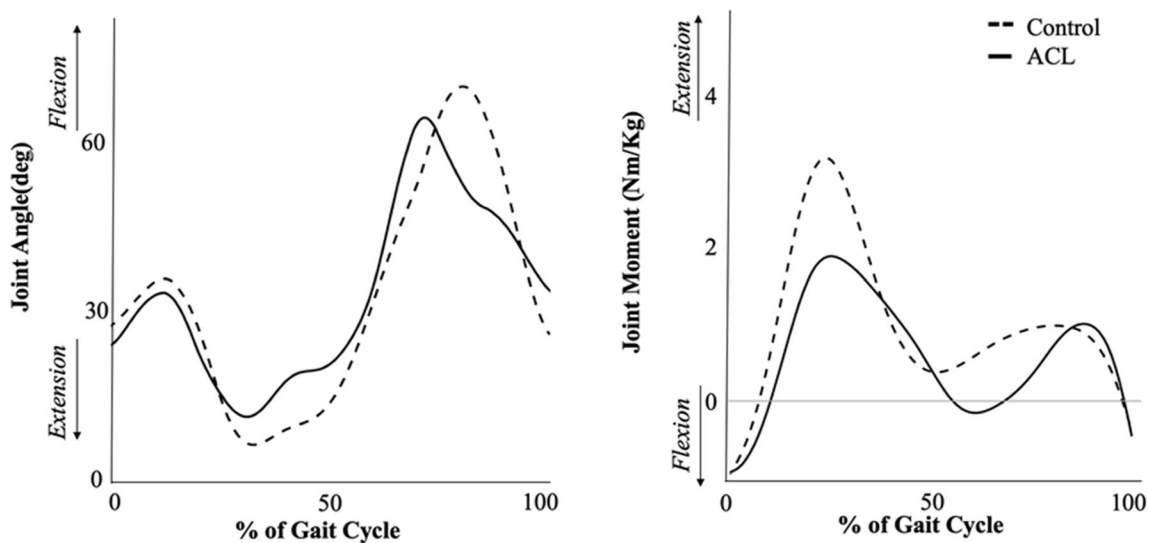


Fig. 2 Sagittal plane knee angle (left) and sagittal plane knee moments (right) during the running gait cycle

[17, 18, 25, 26]. Furthermore, regenerative and translational engineering aims to enhance the clinical effectiveness of the decisions and strategies used to bioengineer new tissues to meet the material and mechanical requirements [19]. While the bioengineered tissue needs to exhibit similar material properties such as tissue volume, distribution, and density, the materials are functionally useless if they cannot withstand the mechanical loading [17, 27]. Traditionally, mechanical testing serves as the gold standard for evaluating the structural efficacy of bioengineered materials; however, this testing does not always replicate the loading behavior during dynamic tasks. Here, computational modeling can provide greater insight into the dynamic loading of the ACL and the mechanisms that lead to failure that can enhance the design and optimization of new bioengineered materials and improve its structural and functional performance.

This study analyzed ACL loading during running despite most ACL tears occurring during cutting or jumping tasks. This group was selected because they had undergone reconstruction surgery 6 months prior to their participation in the study. Although these individuals were cleared to participate in sport tasks such as cutting and landing, it is understood that post-ACLR individuals often still suffer from neuromuscular deficits 6 months post-ACLR placing them at higher risk for reinjury. Running provided a safe compromise since the physical demands required to execute the task would produce differences in movement patterns that would impact ACL loading, but the forces in the ACL would not approach injurious levels. This study was successful in producing differences in movement patterns between the healthy controls and post-ACLR individuals that revealed information about how these differences in movement patterns impact ACL loading.

ACL loading was higher in the post-ACLR individuals than the healthy controls, but the difference was not

significant. The fact that differences in ACL forces were not significant despite the post-ACLR individuals producing significantly lower knee flexion and reduced knee extensor moments suggests that the adopted movement strategies by the post-ACLR individuals may not increase loading on the ACL to the injurious levels that has been previously believed.

Given that the loading is not significantly higher than in healthy controls indicates that secondary ACL tears may be attributed to the structural integrity of the implanted grafts. Thus, future work should continue to engineer novel materials to withstand the variable loading conditions.

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Data Availability Not applicable.

Code Availability Not applicable.

Declarations

Conflict of Interest The authors declare no competing interests.

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